Type MS · What the operating system does, and system vs application software

Answer 1 of 6 · Question 5 part e · 2 marks

2023 JUN · Paper 01 · extract 1 of 8 · find part e in this mark scheme extract

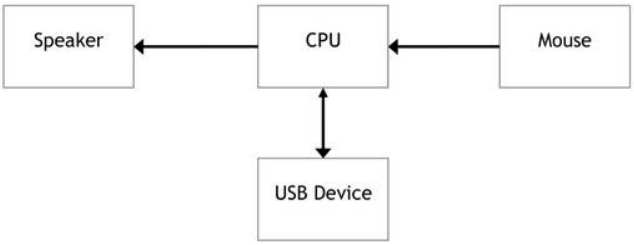
<table>
<tr>
<td>5(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Req.</th><th>Example</th></tr>
<tr><td>Input</td><td>1. Enter the direction the bee will face (1)
2. Enter the number of steps that the bee will move (1)</td></tr>
<tr><td>Process</td><td>Calculate the path the bee will move along to its new position</td></tr>
<tr><td>Output</td><td>3. The bee appears in its new position (1)
4. The bee flies to its new position (1)
5. The bee faces its new direction (1)</td></tr>
</table>
</td>
<td>Allow direct instructions as input e.g. ‘Turn East’ and consequences for output e.g. ‘Bee turns to face East’ if clear.

The input/output do not have to be directly linked. E.g. Input Turn West, Output Has Moved 20 steps</td>
<td>2</td>
</tr>
</table>

Type MS · What the operating system does, and system vs application software

Answer 1 of 6 · Question 5 part e · 2 marks

2023 JUN · Paper 01 · extract 2 of 8 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5b(i)	Award one mark for each: • Line between Mouse and CPU with arrow at CPU end only (1) • Line between CPU and Speaker with arrow at Speaker end only (1) • Line between CPU and USB Device with arrows on both ends (1) 	Allow two independent back/forth arrows between CPU and USB in place of a single bi-directional arrow. Ignore any additional lines.	3

Type MS · What the operating system does, and system vs application software

Answer 1 of 6 · Question 5 part e · 2 marks

2023 JUN · Paper 01 · extract 3 of 8 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(ii)	**The only correct answer is B** ***A*** *is not correct because a* ***firewall*** *is an application that is loaded after the device is booted* ***C*** *is not correct because* ***the operating system*** *is loaded after the device is booted* ***D*** *is not correct because* ***the user interface*** *is loaded after the device is booted*		1

Question Number	Answer	Additional Guidance	Mark
5(b)(iii)	Award up to **two** marks for a linked explanation, such as: • Cache makes up for the difference in speed of two devices (1) because it is very high-speed memory (1) • Cache is high-speed memory (1) which acts as a buffer between a faster and a slower device (1) • It speeds up processing (1) because active/commonly used instructions and data are stored nearer to the CPU / the CPU doesn't have to wait while instructions/data are fetched from RAM (1)		2

Type MS · What the operating system does, and system vs application software

Answer 1 of 6 · Question 5 part e · 2 marks

2023 JUN · Paper 01 · extract 4 of 8 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)	Award up to **two** marks for a linked description, such as: • Uses a sensor (1) to check the water level in the machine (1) • Uses a thermometer (1) to check the water temperature (1) • Uses a timer (1) to control the time required for the wash cycle/rinse/drain/spin (1) • Uses a motor (1) to rotate the drum (1)	The first mark is for identification and the second is for the linked description. It is possible to gain one mark for a description without identifying the sensor/actuator.	2

Type MS · What the operating system does, and system vs application software

Answer 1 of 6 · Question 5 part e · 2 marks

2023 JUN · Paper 01 · extract 5 of 8 · find part e in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award one mark for each two correct rows.

<table>
<tr><th>A</th><th>B</th><th>C</th><th>A AND C</th><th>(A AND C) OR B</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td></td><td>3</td></tr>
</table>

Type MS · What the operating system does, and system vs application software

Answer 1 of 6 · Question 5 part e · 2 marks

2023 JUN · Paper 01 · extract 6 of 8 · find part e in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(ii)</td><td>Award one mark for each of:

• M OR P (1)
• NOT C (1)
• <expr1> AND <expr2> (1)

Example statements:

(M OR P) AND (NOT C)
(M OR P) AND NOT C
(M AND NOT C) OR (P AND NOT C)
M AND NOT C OR P AND NOT C

For information only:

<table>
<tr><th>M</th><th>P</th><th>C</th><th>M OR P</th><th>NOT C</th><th>(M OR P) AND (NOT C)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
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<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td><td>Third bullet awarded even if first two are not

Order of precedence is NOT, AND, OR

Check brackets carefully, if provided</td><td>3</td></tr>
</table>

Type MS · What the operating system does, and system vs application software

Answer 1 of 6 · Question 5 part e · 2 marks

2023 JUN · Paper 01 · extract 7 of 8 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. The operating system uses a scheduler to control processes (1) 2. The operating system holds processes in a queue (1) 3. Some processes may be given higher priorities than others (1) 4. Each process gains accesses to the CPU for a short time / time slice to execute (1) 5. Processes are swapped to/from (the queue / CPU)	Do not award responses about memory management, as the question is about sharing the CPU only	2

Type MS · What the operating system does, and system vs application software

Answer 1 of 6 · Question 5 part e · 2 marks

2023 JUN · Paper 01 · extract 8 of 8 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What the operating system does, and system vs application software

Answer 2 of 6 · Question 1 part f · 1 mark

2019 JUN · Paper 01 · extract 1 of 3 · find part f in this mark scheme extract

1(a)	Award **one** from: Input, process, output in this order – **3 marks** Input, process, output in any order – **2 marks** At least one of input, process, output – **1 mark**		3

Question Number	Answer	Additional Guidance	Mark
1(b)	B: Central processing unit **The only correct answer is B** **A** is not correct as this is memory only **C** is not correct as this is memory only **D** is not correct as this is used only for transporting memory addresses		1

Question Number	Answer	Additional Guidance	Mark
1(c)	• Data (1) • Instructions (1)		2

Question Number	Answer	Additional Guidance	Mark
1(d)	Any **two** from: 1. Virtual memory (VM) is used when RAM becomes full (1) (to hold all programs and data). 2. Virtual memory is used as (an extension to) main memory/RAM / works like RAM. (1) 3. Virtual memory is stored/created on (internal) secondary storage/HDD/SSD. (1) 4. Virtual memory is used as temporary storage. (1) 5. Instructions and data not currently being used are transferred from RAM to VM/HDD. (1) 6. When needed again, instructions and data are transferred back to RAM. (1)		2

Type MS · What the operating system does, and system vs application software

Answer 2 of 6 · Question 1 part f · 1 mark

2019 JUN · Paper 01 · extract 2 of 3 · find part f in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(e)	Any **two** from: 1. The disc surface is made out of polycarbonate. (1) 2. A laser (beam) is used. (1) 3. Data is stored as a series of tiny grooves arranged in tracks on the surface of the disc. (1) 4. Bumps/lands/flats/troughs/pits/dips imprinted/burnt on the disc. (1) 5. Bumps and flats (etc.) represent 1 and 0 (bits) / the reflection (of the laser) represents 1 and 0 (bits). (1)		2

Question Number	Answer	Additional Guidance	Mark
1(f)	D: System software **The only correct answer is D** **A** is not correct because this category is purely for software that is used for security purposes **B** is not correct because this category is designed for end user tasks rather than for managing the computer itself **C** is not correct because this software is used to translate a high level program into machine code		1

Type MS · What the operating system does, and system vs application software

Answer 2 of 6 · Question 1 part f · 1 mark

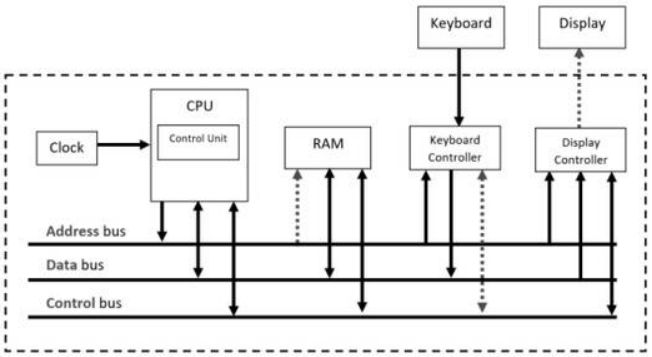
2019 JUN · Paper 01 · extract 3 of 3 · find part f in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What the operating system does, and system vs application software

Answer 3 of 6 · Question 3 part d · 4 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

3(a)	 • **Address bus** labelled correctly (1) • **Data bus** labelled correctly (1) • **Control bus** labelled correctly (1) • Arrow from address bus to RAM (1) • Bi-Directional arrows from control bus to keyboard controller (1) • Arrow points from display controller to display (1)		6

Question Number	Answer	Additional Guidance	Mark
3(b)	Award **two** marks for a linked explanation such as: The cache stores frequently used data/instructions (1) so the processor does not have to wait because cache is checked before main memory / because RAM is further away from the processor (1). It speeds up processing (1) because cache is a faster type of memory (1).		2

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Type MS · What the operating system does, and system vs application software

Answer 3 of 6 · Question 3 part d · 4 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	<table><tr><th>CPU1</th><th>CPU2</th></tr><tr><td>5 * 2 = 10</td><td>4 * 3 = 12 (1)</td></tr><tr><td>10 + 12 = 22 (1)</td><td>24 / 4 = 6 (1)</td></tr><tr><td>22 - 6 = 16 (1)</td><td></td></tr></table>	The second row expressions can be in either column 22 – 6 = 16 May appear in either CPU in the final row	4

Question Number	Answer	Additional Guidance	Mark
3(d)	Any four from: • The OS checks whether sufficient space is available on the storage (media) for the file (1). • The file is broken into blocks (1) • The blocks are stored in spaces that are large enough (1) • OS looks for (fat/ntfs)/ reserves (linux) (groups of) contiguous blocks (1) • Blocks can reside anywhere on the storage (1). • The OS updates the file allocation table (with the start location and sequence number of each block) (1) • OS sets hard links to files (1) ntfs • OS updates journal/MFT (1) ntfs/linux • Metadata about the file such as read/write permissions, date created and last accessed is separately stored (1)		4

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Type MS · What the operating system does, and system vs application software

Answer 3 of 6 · Question 3 part d · 4 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What the operating system does, and system vs application software

Answer 4 of 6 · Question 3 part c · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

3(a)	Award up to **two** marks for a linked explanation, such as: • If the data is incomplete/inaccurate (1) the answers from the model might not be right (1) • If the assumptions the model is based on are inaccurate (1) the answers from the model might be incomplete/inaccurate (1)	Award one mark for identifying a problem and one mark for a reason There must be a clear cause and consequence/effect	2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for any of: • Pixels per inch/ppi (1) • Pixels per centimetre/ppc (1) • Dots per inch/dpi (1)	Ignore units cm^2 $inches^2$	1

Type MS · What the operating system does, and system vs application software

Answer 4 of 6 · Question 3 part c · 2 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

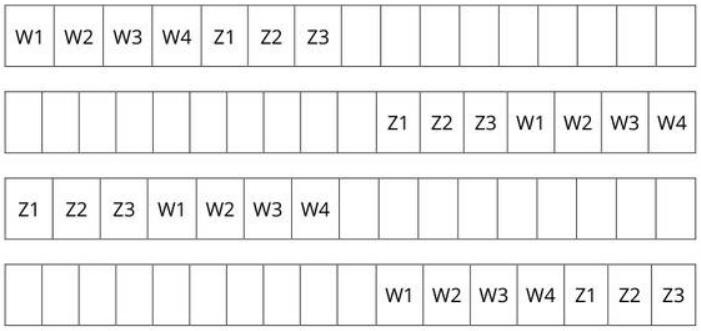
Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	Award **two** marks for a linked description such as: • Each pixel/colour in the image (1) will require more bits to store (1)	Giving for formula file_size = width*height*colour_depth on its own is not enough	2

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked description such as: • System software is used to maintain/run the operation of the system/hardware (1) whereas application software allows the user to carry out a task (1)	For both marks both categories must be addressed Allow clear examples	2

Type MS · What the operating system does, and system vs application software

Answer 4 of 6 · Question 3 part c · 2 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

Question number	Answer	Additional guidance	Mark
3(d)	Award **one** mark for each of: • All file blocks of each file are in numeric sequence/one after another in numeric order (1) • All blank space is together at the front/at the end (1) Examples: 	Note: Order of letter and number does not matter. W1 or 1W are equally acceptable	2

Type MS · What the operating system does, and system vs application software

Answer 4 of 6 · Question 3 part c · 2 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)	Award up to **two** marks for a linked explanation such as: • The file size will not be decreased/will increase (1) because there are very few/no repeating patterns / because a run length of one would be added to each digit (1)		2

Question Number	Answer	Additional Guidance	Mark
3(f)	Award **one** mark for each correct cell:		4

Task	Translator
A guessing game that can be used on different computing platforms	Interpreter (1)
A screen driver for a new smartphone	Assembler (1)
A new version of a spreadsheet program for sale next year	Compiler (1)
Control software for an embedded system inside a new washing machine	Assembler (1)

Type MS · What the operating system does, and system vs application software

Answer 4 of 6 · Question 3 part c · 2 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What the operating system does, and system vs application software

Answer 5 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

2(a)(i)	Award **one** mark for any of the following: • To perform end **user** related tasks (1) • To help the user be more productive/creative/communicative (1)	Do not award trade names or generic categories of software Allow software examples such as word processor if it is clear that it is an end-user task. Do not award responses related to tasks that could be performed by pen/paper.	1
2(a)(ii)	Award **one** mark for any of the following up to a maximum of **three** marks: • Process management (1) • Memory management/virtual memory (1) • File management (1) • Peripheral/hardware management (1) • Network management (1) • Security management (1) • User management (1)	Allow equivalence to management such as 'scheduling processes', 'Virtual memory organisation', but do not award terms in isolation. Do not award marks for types of OS such as multitasking Do not award UI as given in question.	3

Type MS · What the operating system does, and system vs application software

Answer 5 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

2(b)(i)	Award up to **two** marks for a linked explanation such as: • RAM is volatile/its contents will be lost when the power is switched off (1) ROM (is permanent) is required to store the BIOS/firmware/to boot up the computer (1) • The contents of ROM cannot be changed (1) RAM is required for actively executing programs and data (1)	Both RAM and ROM must be addressed for 2 marks.	2
2(b)(ii)	Award **one** mark for any of the following up to a maximum of **three** marks: • A platter is used (1) • (The platter) is divided into tracks / sectors (1) • A read/write arm/head is moved (to the track required) (1) • A read/write head reads the values (from the surface) (1) • One polarity represents 1 and other polarity represents 0 (1)	Accept North/South for polarity and representation of binary state.	3

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Type MS · What the operating system does, and system vs application software

Answer 5 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 01 · extract 3 of 4 · find part a.i in this mark scheme extract

<table>
<tr>
<td>2(c)</td>
<td>Award one mark for each correct row:
<table>
<tr><th>Stage</th><th>Sequence Number</th></tr>
<tr><td>The instruction is placed onto the data bus</td><td>2</td></tr>
<tr><td>The Arithmetic and Logic Unit (ALU) performs a calculation</td><td>4</td></tr>
<tr><td>The address of the next instruction is placed onto the address bus</td><td>1</td></tr>
<tr><td>The instruction is decoded by the control unit</td><td>3</td></tr>
</table>
</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>2(d)</td>
<td>The only correct answer is A

B is not correct because the control bus is bidirectional
C is not correct because the system bus is a collective term that contains both unidirectional and bidirectional buses
D is not correct because the data bus is bidirectional</td>
<td></td>
<td>1</td>
</tr>
<tr>
<td colspan="2"></td>
<td>Total for question 2</td>
<td>14</td>
</tr>
</table>

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Type MS · What the operating system does, and system vs application software

Answer 5 of 6 · Question 2 part a.i · 1 mark

2024 JUN · Paper 01 · extract 4 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What the operating system does, and system vs application software

Answer 6 of 6 · Question 2 part a.ii · 3 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

Number			
2(a)(i)	Award **one** mark for any of the following: • To perform end **user** related tasks (1) • To help the user be more productive/creative/communicative (1)	Do not award trade names or generic categories of software Allow software examples such as word processor if it is clear that it is an end-user task. Do not award responses related to tasks that could be performed by pen/paper.	1
2(a)(ii)	Award **one** mark for any of the following up to a maximum of **three** marks: • Process management (1) • Memory management/virtual memory (1) • File management (1) • Peripheral/hardware management (1) • Network management (1) • Security management (1) • User management (1)	Allow equivalence to management such as 'scheduling processes', 'Virtual memory organisation', but do not award terms in isolation. Do not award marks for types of OS such as multitasking Do not award UI as given in question.	3

Type MS · What the operating system does, and system vs application software

Answer 6 of 6 · Question 2 part a.ii · 3 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part a.ii in this mark scheme extract

2(b)(i)	Award up to **two** marks for a linked explanation such as: • RAM is volatile/its contents will be lost when the power is switched off (1) ROM (is permanent) is required to store the BIOS/firmware/to boot up the computer (1) • The contents of ROM cannot be changed (1) RAM is required for actively executing programs and data (1)	Both RAM and ROM must be addressed for 2 marks.	2
2(b)(ii)	Award **one** mark for any of the following up to a maximum of **three** marks: • A platter is used (1) • (The platter) is divided into tracks / sectors (1) • A read/write arm/head is moved (to the track required) (1) • A read/write head reads the values (from the surface) (1) • One polarity represents 1 and other polarity represents 0 (1)	Accept North/South for polarity and representation of binary state.	3

Type MS · What the operating system does, and system vs application software

Answer 6 of 6 · Question 2 part a.ii · 3 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part a.ii in this mark scheme extract

<table>
<tr>
<td>2(c)</td>
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<table>
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<tr><td>The Arithmetic and Logic Unit (ALU) performs a calculation</td><td>4</td></tr>
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</table>
</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>2(d)</td>
<td>The only correct answer is A

B is not correct because the control bus is bidirectional
C is not correct because the system bus is a collective term that contains both unidirectional and bidirectional buses
D is not correct because the data bus is bidirectional</td>
<td></td>
<td>1</td>
</tr>
<tr>
<td colspan="2"></td>
<td>Total for question 2</td>
<td>14</td>
</tr>
</table>

Type MS · What the operating system does, and system vs application software

Answer 6 of 6 · Question 2 part a.ii · 3 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark